

International Institute of Information Technology, Bangalore
CSE 511 Algorithms: Practice Problem

Description: You are given two integer arrays `speed` and `efficiency`, each of length n , where `speed[i]` and `efficiency[i]` denote the speed and efficiency of the i -th engineer. You are also given an integer k .

You must choose *at most* k engineers out of the n engineers to form a team. The *performance* of a team is defined as:

$$\text{performance} = \left(\sum_{i \in \text{team}} \text{speed}[i] \right) \times \left(\min_{i \in \text{team}} \text{efficiency}[i] \right).$$

Return the *maximum* performance of a team. Since the answer may be large, return it modulo $10^9 + 7$.

Input:

- The first line contains two integers n and k .
- The next n lines each contain two integers: `speed[i]` and `efficiency[i]`.

Output: Print a single integer — the maximum performance modulo $10^9 + 7$.

Example 1:

Input :

```
6 2
2 5
10 4
3 3
1 9
5 7
8 2
```

Output :

```
60
```

Explanation: Choose engineer with `speed` = 10, `efficiency` = 4 and engineer with `speed` = 5, `efficiency` = 7:

$$\text{performance} = (10 + 5) \times \min(4, 7) = 15 \times 4 = 60.$$

This is the maximum possible with $k = 2$.

Example 2:

Input :

```
6 3
2 5
10 4
```

```
3 3
1 9
5 7
8 2
```

Output:
68

Explanation: Choose engineers with (speed, efficiency) pairs (1, 9), (5, 7), and (10, 4):

$$\text{performance} = (1 + 5 + 10) \times \min(9, 7, 4) = 16 \times 4 = 64.$$

Not optimal. The optimal team is (1, 9), (2, 5), and (10, 4):

$$\text{performance} = (1 + 2 + 10) \times \min(9, 5, 4) = 13 \times 4 = 52.$$

The true maximum is achieved by (2, 5), (5, 7), (10, 4):

$$\text{performance} = (2 + 5 + 10) \times \min(5, 7, 4) = 17 \times 4 = 68.$$

Example 3:

Input:

```
6 4
2 5
10 4
3 3
1 9
5 7
8 2
```

Output:
72

Explanation: Choose engineers (1, 9), (2, 5), (5, 7), and (10, 4):

$$\text{performance} = (1 + 2 + 5 + 10) \times \min(9, 5, 7, 4) = 18 \times 4 = 72.$$

Constraints:

- $1 \leq k \leq n \leq 10^5$
- $1 \leq \text{speed}[i] \leq 10^5$
- $1 \leq \text{efficiency}[i] \leq 10^8$